AGENTIC AI LABS • PROJECT 04

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E-commerce Delivery Performance

Data-Driven Insights & Business Recommendations

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Project Overview

Project Purpose

This project aims to analyse e-commerce delivery performance and identify opportunities to improve the customer experience.

Business Problem

Delivery delays and inconsistent service impact customer satisfaction and increase operational overheads.

Project Objective

Analyse delivery performance data, identify potential delivery issues, and recommend actionable improvements.

EXPECTED OUTCOME: A clear business optimization strategy supported by data insights and operational efficiency.

SYSTEM ARCHITECTURE

Structured Output Extraction Pipeline

01

INPUT DOCUMENTS

Invoices, Résumés and Job Posts

02

DATA EXTRACTION

Extract names, dates, amounts and emails

03

VALIDATION

Check schema, field types and required values

04

FINAL OUTPUT

Valid JSON or retry/manual review

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DATA SCHEMA & VALIDATION

Ensuring Accurate and Reliable Structured Data

FIELD	DATA TYPE	RULES
Name	`String`	Text Value
Date	`Date`	Valid date
Amount	`Number`	Numeric value
Email	`String`	Valid email format

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DATA SCHEMA & VALIDATION

Ensuring Accurate and Reliable Structured Data

Type Check

Confirm each value matches its expected data type.

Required Fields

Check whether mandatory fields are present.

Error Handling

Retry invalid output or send it for manual review.

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ERROR HANDLING & RECOVERY

Reliable Validation with a Clear Failure Path

STEP 01

Extracted Output

Receive structured data from the extraction step.

STEP 02

Validate Output

Check schema, field types and required values.

VALID
Accept and save output

INVALID
Record validation errors

STEP 03

Retry or Manual Review

Retry failed extraction or route for human review.

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DATASET & TEST SAMPLES

Testing Extraction Across Different Document Types

Invoices

Names, dates and amounts

Résumés

Names, emails and skills

Job Posts

Job titles and requirements

No-Field Documents

Documents with no target fields

50+

TEST SAMPLES

Minimum evaluation target specified in the project brief.

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EXTRACTION ACCURACY & EVALUATION

Measuring the Reliability of Structured Data Extraction

Field-level Accuracy

Correctly extracted fields compared with expected values.

Schema Validity

Percentage of outputs that pass schema validation.

Empty-Field Handling

Correct handling of documents with no target fields.

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EXTRACTION ACCURACY & EVALUATION

Measuring the Reliability of Structured Data Extraction

EVALUATION METRIC

Accuracy Formula

$$\text{Accuracy} = \frac{\text{Correct Field}}{\text{Total Expected Field}} \times 100$$

CURRENT EVALUATION

Results Status

To be measured after testing 50+ samples.

Actual accuracy and validation results will be added after evaluation.

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CONCLUSION & FUTURE IMPROVEMENTS

Project Goals

- Convert unstructured text into structured data.
- Validate outputs against a typed schema.
- Retry invalid outputs or route them for review.
- Evaluate extraction accuracy on 50+ samples.

Future Improvements

- Improve extraction accuracy.
- Support additional document formats.
- Strengthen error handling and validation.
- Expand the test dataset.

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THANK YOU

Structured Output Extraction Pipeline